Royce Moon

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Experience

Enclave Intelligence | *Co-Founder & Managing Partner* Mar 2026 – Present

- Co-founded an AI transformation partner company advising and engineering for institutions, generational firms, and operators. Scaled the firm to a six-figure annualized revenue run rate within five months by advising C-suite teams on AI investment strategy and defining build programs with explicit targets for ROI, revenue growth, cost savings, and operating leverage.
- Built cross-functional enterprise knowledge bases that unify fragmented documents, communications, and operating data through permission-aware retrieval, deployed function-specific agents on top of those systems, and developed proprietary automated red-team infrastructure to test data access, tool use, and multi-step behavior over long horizons.

Xchg Labs | *CTO (Now Advisor)* Nov 2025 – Present

- Built Vex, a first-of-its-kind offensive multi-agentic security framework leveraging formal verification to discover 20 critical remote code execution zero-days in local AI inference frameworks and NVIDIA container software.
- Co-developed Meridian, a cyber reasoning system powered by an interprocedural taint engine and used it to discover 5 CVEs in dnsmasq (CVE-2026-4893, CVE-2026-4892, CVE-2026-4891, CVE-2026-4890, CVE-2026-2291) and 6 CVEs in Oracle REST Data Services (CVE-2026-46840 which has a maximum CVSS of 10.0, CVE-2026-46829, CVE-2026-46841, CVE-2026-46842, CVE-2026-46843, CVE-2026-46830). 50 additional zero-days discovered in router firmwares, web server frameworks, and many other softwares.

Spiral Works | *Founder & CEO* Jul 2024 – Nov 2025

- Founded a venture-backed AI research lab, raising $1M+ pre-seed to build creative language models for open-ended discovery. Built a dataset of 20 million pages of multi-disciplinary scientific papers and corresponding peer reviews and was developing reinforcement learning environments to train frontier judge models for science.
- Led 6 research affiliates to publish 2 papers: first LLM evaluations framework for combinatorial creativity (ICLR 2026 under review) and a LLM system for creative scientific idea generation (ICCC 2026). Team also produced the first formal graph theory for transformational creativity (Best Short Paper Award, ICCC 2026).

University of Pennsylvania | *Undergraduate Researcher, Automatic Speech Recognition* Apr 2024 – Jul 2025

- Architected and built the first heart failure telehealth ASR dataset from 114 long-horizon recordings under the NIH-funded HEAR-HEARTFELT study. Implemented forced alignment using WhisperX and hash-based train/val/test splitting to prevent conversation leakage.
- Validated dataset efficacy by finetuning Whisper large-v3-turbo via LoRA and rank-stabilized LoRA adapters. Empirically validated rank-stabilized scaling, achieving 22.5% word error rate reduction using rank-stabilized LoRA and 15.1% word error rate reduction using LoRA.

AbbVie | *Machine Learning Researcher, New Biological Entities R&D Group* Apr 2022 – May 2023

- Spearheaded the development of a proof-of-concept to show ML applications were possible in peptide mapping classification for antibody drug conjugates. Scaled to production towards the capacity of 8,000+ analysts.
- Increased the antibody drug conjugate development pipeline efficiency by 35%, saved 1000s of man hours for chemistry analysts, and presented research to 300+ research scientists and VPs at an international, internal symposium for engineering a cloud ML pipeline for antibody drug conjugate peptide mapping classification using SVM, CNN, and hierarchical clustering models.

Research Publications

R. Moon, L. R. Varshney, *Containment Verification: AI Safety Guarantees Independent of Alignment*, Second Workshop on Agents in the Wild at the International Conference on Machine Learning (ICML), 2026.

A. Sanyal, S. Schapiro, S. Shashidhar, **R. Moon**, L. R. Varshney, D. Hakkani-Tur, *Spark: A System for Scientifically Creative Idea Generation*, Proceedings of the 16th International Conference on Computational Creativity (ICCC), 2026.

S. Schapiro, S. Shashidhar, A. Gladstone, J. Black, **R. Moon**, D. Hakkani-Tur, L. R. Varshney, *Combinatorial Creativity: A New Frontier in Generalization Abilities*, preprint, 2026.

Education

University of Michigan Graduated August 2025

B.S. in Computer Science *Ann Arbor, MI*

- **Select Coursework:** Applied Linear Algebra, Intro to Artificial Intelligence, Algorithms for ML & Data Science, Formal Verification of Systems Software, Software Engineering, Theory of Complex Systems

University of Illinois, Urbana-Champaign Transferred

B.S. in Computer Science + Philosophy *Champaign, IL*

- **Organizations:** Quant Illinois (Co-Founder, First Head of Research, Advisor)